

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

Team ID: SWTID-2026-5839

Team Leader: Rahul Kumawat

PROJECT FINAL REPORT

DATE: 8 July 2026

1. INTRODUCTION

1.1 Project Overview

The housing market plays a significant role in **economic development, investment planning, and urban growth**. House prices are influenced by several factors such as property size, number of bedrooms and bathrooms, house condition, renovation history, location, and other structural features. Understanding how these factors affect property values is essential for making informed decisions in the real estate sector.

The “Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau” project provides meaningful insights into housing market patterns by examining the relationship between house characteristics and sale prices. The project highlights trends in property values, renovation impact, house age, and structural features, enabling users to better understand the factors that contribute to housing prices.

Through interactive visualizations, the project supports home buyers, sellers, real estate professionals, and researchers in comparing property features, identifying pricing trends, and evaluating the influence of different housing attributes. By presenting housing information in a clear and accessible manner, the project helps improve decision-making in the real estate market.

1.2 Purpose

The purpose of the “Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau” project is to provide valuable insights into housing market trends by examining how different property features influence sale prices.

The project aims to:

- * Help home buyers understand the factors that affect property prices before making purchasing decisions.

- * Assist sellers in estimating competitive selling prices based on property characteristics and market trends.
- * Support real estate professionals in evaluating housing market performance and pricing patterns.
- * Identify the impact of house features such as bedrooms, bathrooms, floor area, house age, renovation status, and overall condition on property values. *
- Examine the relationship between renovation history and house sale prices to understand the value of property improvements.
- * Enable stakeholders to compare housing characteristics across different locations and property categories.
- * Promote informed and data-driven decision-making in the housing market through clear visualization of pricing and property trends.

This project serves as a practical decision-support tool that helps stakeholders better understand housing market behaviour, evaluate property values, and make informed real estate decisions based on key housing features and market trends.

2. IDEATION PHASE

2.1 Problem Statement

Customer Problem Statement Template:

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

This customer problem statement aligns with the project "Visualizing Housing Market Trends: An Analysis of Sale Prices and Features." It highlights the challenges faced by home buyers, sellers, and real estate investors in understanding property prices and market trends. The project analyses housing sale prices along with important property features such as location, size, bedrooms, and bathrooms. Interactive dashboards help users compare properties and identify market patterns. These insights support informed buying, selling, and investment decisions. The project makes housing data easier to understand and improves decision-making through clear visualizations.



Customer Problem Statement Template

Use this template to clearly articulate the customer's problem, the challenges they face, and the impact it has on their decisions and outcomes.

01	02	03	04	05
				
I AM	I'M TRYING TO	BUT	BECAUSE	WHICH MAKES ME FEEL
A home buyer, real estate investor, real estate agent, or property developer looking to understand housing market trends and make informed property decisions.	Analyze housing sale prices, identify the key features influencing property values, compare market trends across locations, and make data-driven buying, selling, or investment decisions using interactive Tableau dashboards.	I struggle to interpret large volumes of housing data, identify meaningful trends, compare property prices accurately, and understand how factors such as location, size, number of bedrooms, bathrooms, year built, and amenities affect sale prices.	Housing data is often scattered across multiple sources, contains missing or inconsistent values, and is difficult to analyze without visualization tools. Traditional reports fail to provide interactive insights, making trend analysis time-consuming and less effective.	Confused, uncertain, and hesitant about making property investment or purchasing decisions, increasing the risk of overpaying, missing profitable opportunities, or making poor real estate investments.

**CUSTOMER PROBLEM SUMMARY**

Customers need a simple and interactive way to explore housing market data, uncover meaningful trends, and understand what truly drives property prices so they can make confident, data-driven real estate decisions.

IMPACT



Poor Decision Making



Financial Risk



Time Consuming



Missed Market Opportunities

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A home buyer, real estate investor, real estate agent, or property developer looking to understand housing market trends and make informed property decisions.	Analyse housing sale prices, identify the key features influencing property values, compare market trends across locations, and make data-driven buying, selling, or investment decisions using interactive Tableau dashboards.	I struggle to interpret large volumes of housing data, identify meaningful trends, compare property prices accurately, and understand how factors such as location, size, number of bedrooms, bathrooms, year built, and amenities affect sale prices.	Housing data is often scattered across multiple sources, contains missing or inconsistent values, and is difficult to analyze without visualization tools. Traditional reports fail to provide interactive insights, making trend analysis time consuming and less effective.	Confused, uncertain, and hesitant about making property investment or purchasing decisions, increasing the risk of overpaying, missing profitable opportunities, or making poor real-estate investments.

2.2 Empathy Map Canvas

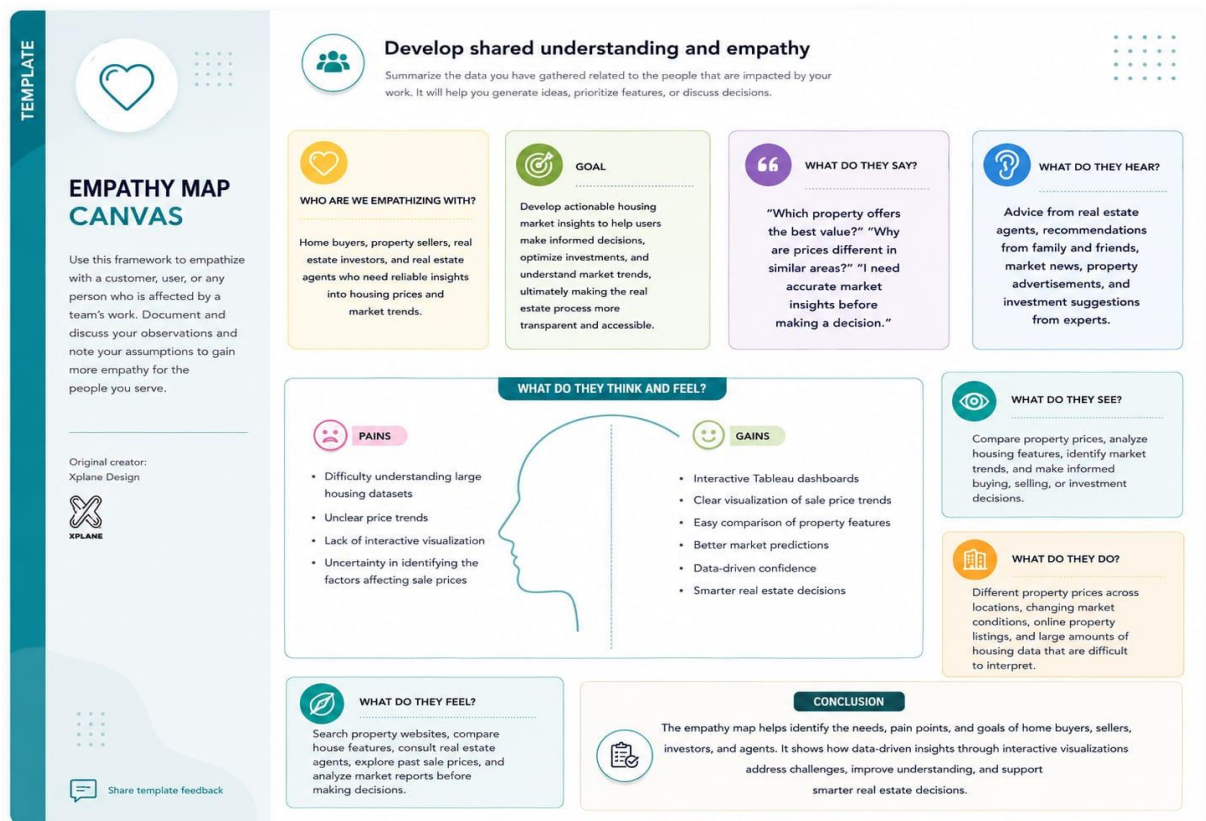
Empathy Map Canvas:

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

These features help home buyers, sellers, investors, and real estate agents make informed decisions, analyse housing market trends, compare property prices, and identify valuable property features.

It is a useful tool to help teams better understand their users.

Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.



2.3 Brainstorming ->

Brainstorm & Idea Prioritization Template:

Brainstorming provides an open environment where team members share ideas for analysing housing market trends using Tableau. It encourages collaboration and creative thinking to identify meaningful visualizations for sale prices and housing features. This process helps develop an interactive dashboard for better analysis and decision-making.

Step-1: Team Gathering, Collaboration and Select the Problem Statement

<u>Team Roster</u> Project Leader Rahul Kumawat <i>Project planning & coordination</i> <i>Task allocation & monitoring</i> <i>Quality review of dashboard</i> <i>Final presentation & reporting</i> Data Analyst & Dashboard Developer Rahul Kumar Meena Data cleaning and processing - Tableau dashboard development - Data visualization and analysis Story creation and insights	<u>[B] Set the goal</u> The goal of this session is to bridge our data visualizations with strategic deployment. We aim to finalize our data calculations (such as fixing the age distribution metrics), refine our dashboard user experience, and successfully map out the integration of our interactive Tableau dashboards into a Flask web application framework.	<u>[problem statement]?</u> (Market Analysis Focus): "How might we transform complex residential housing features—like renovation history, property age, and structural attributes—into an interactive, visual narrative that helps real estate stakeholders optimize pricing strategies and market competitiveness?" (End-User Experience Focus): "How might we build a cohesive web platform that seamlessly embeds advanced data analytics dashboards, enabling users to easily evaluate how age and renovations impact overall property value?" (Technical Deployment Focus): "How might we bridge the gap between heavy business intelligence visualizations (Tableau) and a clean web interface (Flask UI) to deliver a seamless, high-performance analytic tool for real estate decision-makers?"
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Step-2: Brainstorm, Idea Listing and Grouping

Step-2: Brainstorm, Idea Listing and Grouping		
Cluster 1: Interactive Dashboard & Visualization The team proposed developing an interactive Tableau dashboard that allows users to Explore Housing market trends Visualizations Ideas included: - Develop separate geographical maps for 2019 and 2020 Housing market trends. - Create interactive bar charts to compare state-wise housing selling. - Design regional comparison charts to identify differences among Indian regions. - Build a responsive dashboard combining all worksheets in a single interface.	Cluster 2: Dynamic User Interaction The second discussion focused on improving user interaction and dashboard flexibility. Ideas included: - Add Year, Month, Region, and State filters for dynamic exploration. - Create Top N and Bottom N parameter-based visualizations. - Allow dashboard filters to update all worksheets simultaneously. - Enable users to compare housing trends usage across different time periods	Cluster 3: Business Insights & Decision Support The third group focused on generating meaningful business insights. Ideas included: - Analyse housing demand before and during the COVID-19 lockdown. - Identify states with the highest and lowest housing selling. - Compare housing selling across Eastern, Western, Northern, Southern, and North-Eastern regions. - Create a Tableau Story to explain key findings in a logical sequence

Step-3: Idea Prioritization

Step-3: Idea Prioritization

After evaluating every idea based on Importance and Feasibility, the team selected the following priorities.

High Importance / Low Feasibility (Strategic Long-Term Ideas)

- Integrate live Housing data through APIs.
- Build AI-based housing modals prediction models.
- Add real-time forecasting and anomaly detection.
- Develop automated housing trends market for Builders.

High Importance / High Feasibility (Top Priorities)

- Develop interactive Tableau dashboards with multiple visualizations.
 - Build separate geographical maps.
 - Implement dynamic Year, Month, Region, and State filters.
 - Create Top N and Bottom N parameter-based analysis.
- Design regional comparison charts for better understanding of Housing Market trends.

Low Importance / Low Feasibility (Discarded Ideas)

- *Develop a mobile application for dashboard access during the first phase.*
- *Integrate ID-based real-time data.*
- *Build 3D visualization dashboards.*
- *Implement voice-controlled dashboard interaction.*

Low Importance / High Feasibility (Quick Wins)

- *Improve dashboard colour themes and formatting Add informative tooltips to every visualization.*
- *Include dashboard legends and descriptive titles. Enhance navigation using Tableau Story Points.*

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map



3.2 Solution Requirement

Functional Requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Collection & Extraction from Database	The Dataset was downloaded from the Kaggle website , Extracted and then loaded the dataset into Tableau
FR-2	Data Preparation	Data Cleaning and transform data Field Renaming and Final Formatting Prepare data for visualization

FR-3	Data Visualization	Count of Transformed Housing Data Average Sale Prices Area of House from Basement (in Sqft) Total Sales by Years Since Renovation Distribution of House Age by Renovation Status House Age Distribution by Number of Bathrooms, Bedrooms, and Floors
FR-4	Dashboard Visualization and Story	Comprehensive House Price Analysis Dashboard Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau Story
FR-5	Data Performance Testing	Check Number in Visualization and Graphs Utilization of Filters
FR-6	Publishing	Publish Proper Project File in Tableau

Non-functional Requirements:

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Visualization and Dashboard should be easy to understand with simple charts and filters.
NFR-2	Security	Dataset should remain protected and only authorized users can modify the dashboard and Visualization.
NFR-3	Reliability	Visualization and Dashboard should display accurate results without data inconsistency.
NFR-4	Performance	Visualization and Dashboard should load visualizations quickly and respond smoothly to filters.

NFR-5	Availability	Visualization and Dashboard should be available whenever users need to analyse housing data.
NFR-6	Scalability	Dashboard should support additional housing records and new visualizations in the future.

3.3 Data Flow Diagram

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.



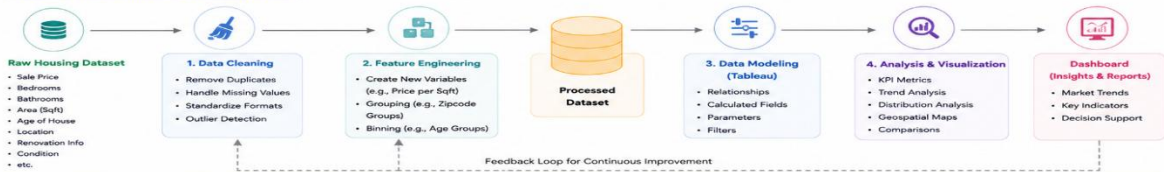
Data Flow Diagrams

A. LEVEL 1 DFD (PROJECT OVERVIEW)

This diagram shows how raw housing data is collected, processed in Tableau, and transformed into meaningful insights that help understand and improve housing market trends.



B. LEVEL 2 DFD (DATA PROCESSING IN DETAIL)



DASHBOARD VISUALIZATIONS



User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Student (Analyst)	Data Collection & Extraction from Database	USN-1	As a Student, I can dataset was downloaded from the Kaggle website, Extracted and then loaded the dataset into tableau.	Data set Downloaded and Uploaded successfully without errors.	High	Sprint-1
Student (Analyst)	Data Preparation	USN-2	As a Student, I can Data Cleaning and transform data Field Renaming and Final Formatting Prepare data for visualization.	Cleaned dataset is ready for Visualization.	High	Sprint-1
Student (Analyst)	Data Visualization	USN-3	As a Student, I can view the Count of Transformed Housing Data and the Average Sale Prices.	Visualizations Correctly Displays the Count of Transformed Housing Data and the Average Sale Prices.	High	Sprint-1

Student (Analyst)	Data Visualization	USN-4	As a Student, I can Area of House from Basement (in sqft.) and Total Sales by Years Since Renovation.	Visualizations update correctly based on the dataset.	High	Sprint-1
Student (Analyst)	Dashboard Visualization and Story	USN-5	As a Student, I can Comprehensive House Price Analysis Dashboard Visualizing Market Trends: An Analysis of Sale Prices and Features using Tableau Story	Dashboard and Story update correctly based on the dataset.	High	Sprint-1
Student (Analyst)	Data Performance Testing	USN-6	As a Student, I can Check Number in Visualization and Graphs Utilization of Filters	Selected filters update all Visualization charts and Utilization Filters correctly.	High	Sprint-2
Student (Analyst)	Trend Analysis	USN-7	As a Student, I can compare sale prices with house features to identify market trends.	Visualization and Dashboard provide meaningful comparisons and insights.	Medium	Sprint-2
Student (Analyst)	Renovation Analysis	USN-8	As a Student, I can analyse total sales by years since renovation.	Dashboard Visualization accurately displays renovation-related sales trends.	Medium	Sprint-2
Student (Analyst)	Report Analysis	USN-9	As a Student, I can use dashboard Visualization insights to support housing market decisions.	Dashboard provides clear and reliable information for decision-making.	Medium	Sprint-2
Student (Analyst)	Publishing	USN-10	Publish Proper Project File in Tableau.	Publish the Project successfully without errors.	Medium	Sprint-2

3.4 Technology Stack

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Guidelines:

1. Include all the processes (As an application Logic/ Technology Block)
2. Provide infrastructural demarcation (Local)
3. Indicate external interfaces (if any)
4. Indicate data storage and components/services
5. Indicate Interface to machine learning models (if applicable)

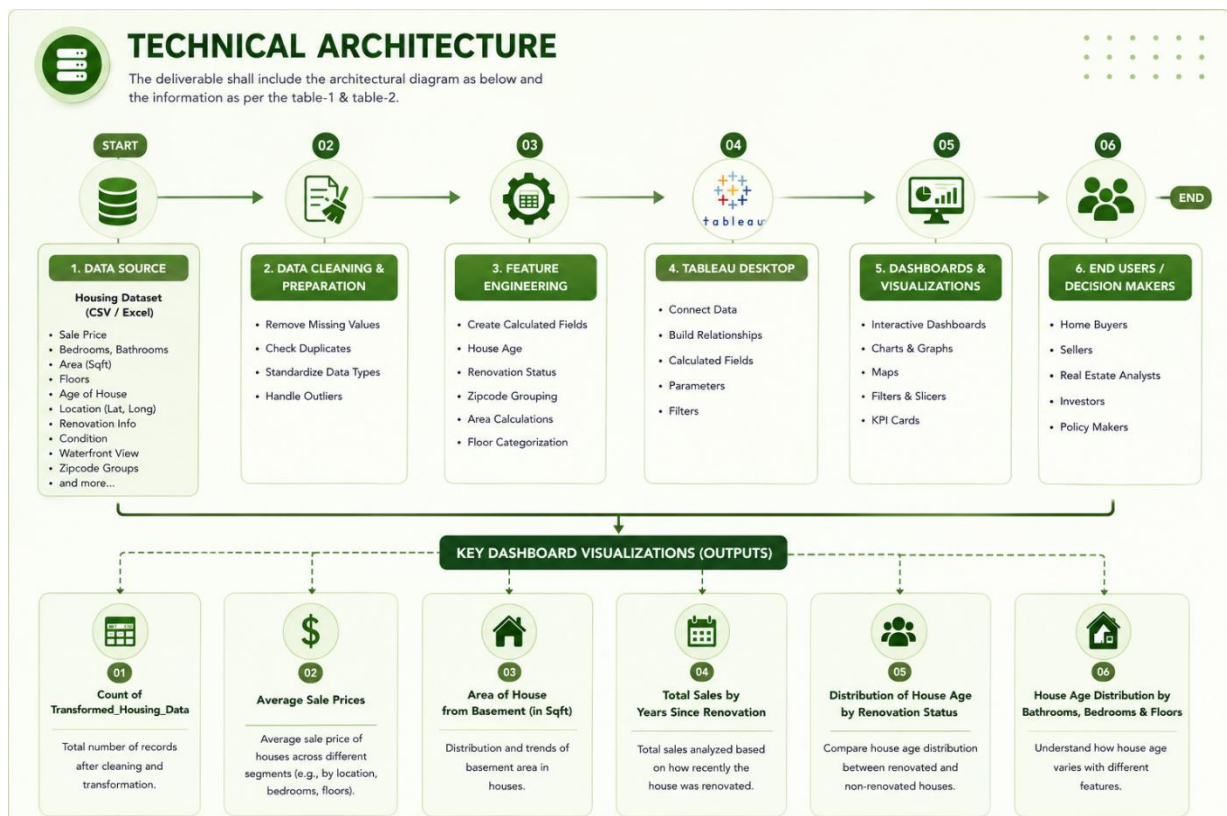


Table-1: Components & Technologies:

S.No	Component	Description	Technology
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1.	User Interface	Interactive dashboard used to explore housing market trends.	Tableau Dashboard
2.	Application Logic-1	Data cleaning and preprocessing before analysis.	Tableau Filters
3.	Application Logic-2	Feature engineering and calculated fields.	Tableau Calculated Fields
4.	Application Logic-3	Interactive analysis using filters, parameters and charts.	Tableau Desktop
5.	Database	Stores the housing dataset for analysis.	CSV / Excel Dataset
6.	Cloud Database	Not used in this project.	N/A
7.	File Storage	Stores dataset and Tableau workbook.	Local Storage (.csv, .xlsx, .twb/.twbx)
8.	External API-1	Not used.	Not Applicable
9.	External API-2	Not used.	Not Applicable
10.	Machine Learning Model	Machine learning is not implemented in this project.	Not Applicable
11.	Infrastructure (Server / Cloud)	Project developed and executed on a local computer.	Windows PC, Tableau Public

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Dataset prepared using Microsoft Excel	Microsoft Excel
2.	Security Implementations	Read-only project files with local system security	Windows File Permissions

3.	Scalable Architecture	Architecture can easily support additional housing records, dashboard sheets and calculated fields without redesigning the system.	Tableau Public, CSV/Excel Dataset
4.	Availability	The dashboard is available whenever the Tableau workbook and dataset are opened on the local computer. Internet is only required for publishing to Tableau Public (if used).	Tableau Public
5.	Performance	Efficient processing of housing data using Tableau's in-memory engine, filters, calculated fields and optimized visualizations for faster dashboard response.	Tableau

4. PROJECT DESIGN

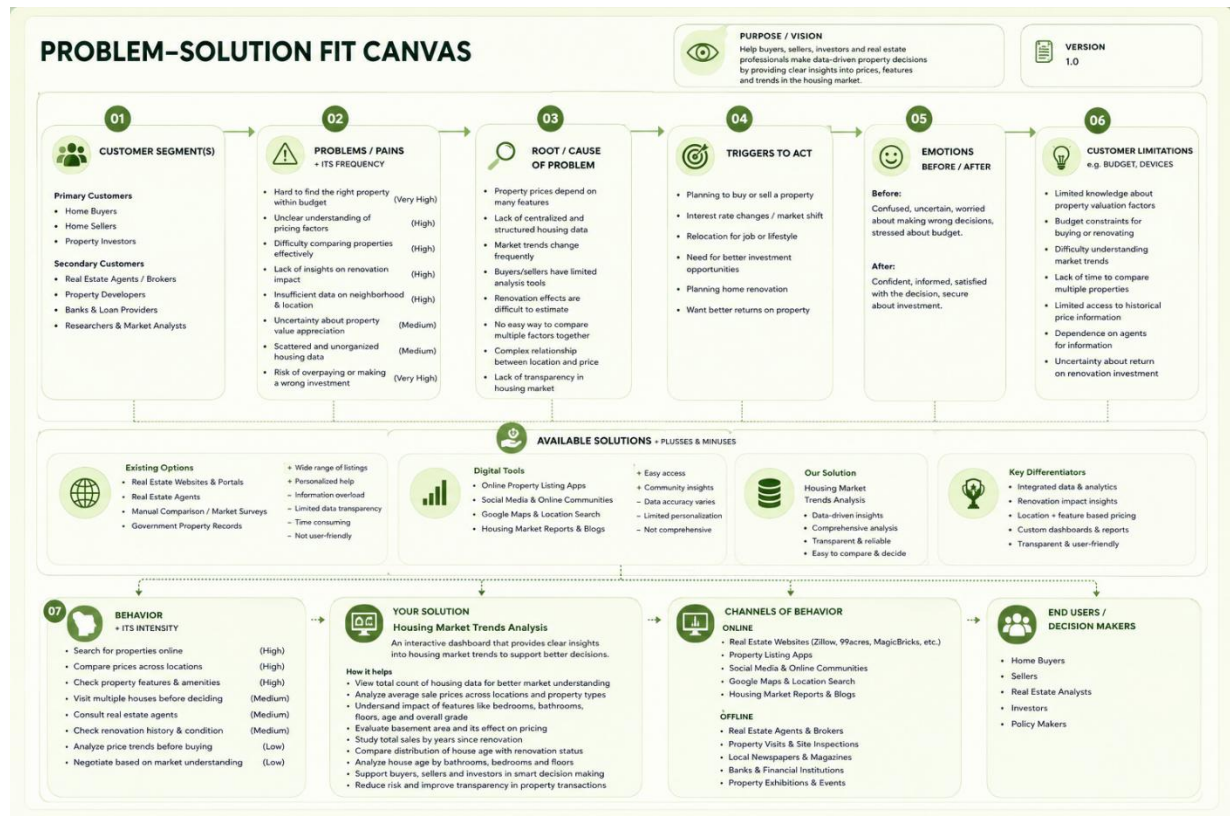
4.1 Problem Solution Fit

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioural patterns and recognize what would work and why

Purpose:

- ☐ Solve complex problems in a way that fits the state of your customers.
- ☐ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behaviour.
- ☐ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ☐ Increase touch-points with your company by finding the right problem-behaviour fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ☐ Understand the existing situation in order to improve it for your target group.



4.2 Proposed Solution:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	The housing market contains a large amount of property data such as sale price, bedrooms, bathrooms, flat area, basement area, house age, renovation status, waterfront view, overall grade. Analysing these attributes manually is difficult and time-consuming, making it challenging for buyers, sellers, and real estate professionals to identify price trends and make informed decisions.
2.	Idea / Solution description	Develop an interactive Tableau dashboard that converts housing data into meaningful visualizations. The dashboard analyses average sale prices, house age, renovation status, basement area, bedrooms, bathrooms, floors, and other property features, enabling users to compare houses easily and make informed decisions based on data-driven insights.

3.	Novelty / Uniqueness	The project combines multiple housing attributes into a single interactive dashboard with filters and visual analytics. Instead of examining raw data, users can instantly identify pricing patterns, renovation impact, and property characteristics through clear visualizations, improving the efficiency of housing market analysis.
4.	Social Impact / Customer Satisfaction	The dashboard helps home buyers, sellers, investors, and real estate agents understand market trends more effectively. It reduces the time required for property comparison, supports transparent pricing decisions, and improves customer confidence through easy-to-understand visual reports.
5.	Business Model (Revenue Model)	The dashboard can be adopted by real estate agencies, property consultants, and housing market analysts as a decision-support tool. Future versions may include premium market reports, subscription-based analytics, or customized dashboards for real estate businesses.
6.	Scalability of the Solution	The solution can easily be expanded by integrating larger housing datasets, additional geographic locations, real-time property information, predictive analytics, and advanced Tableau dashboards without changing the overall project architecture.

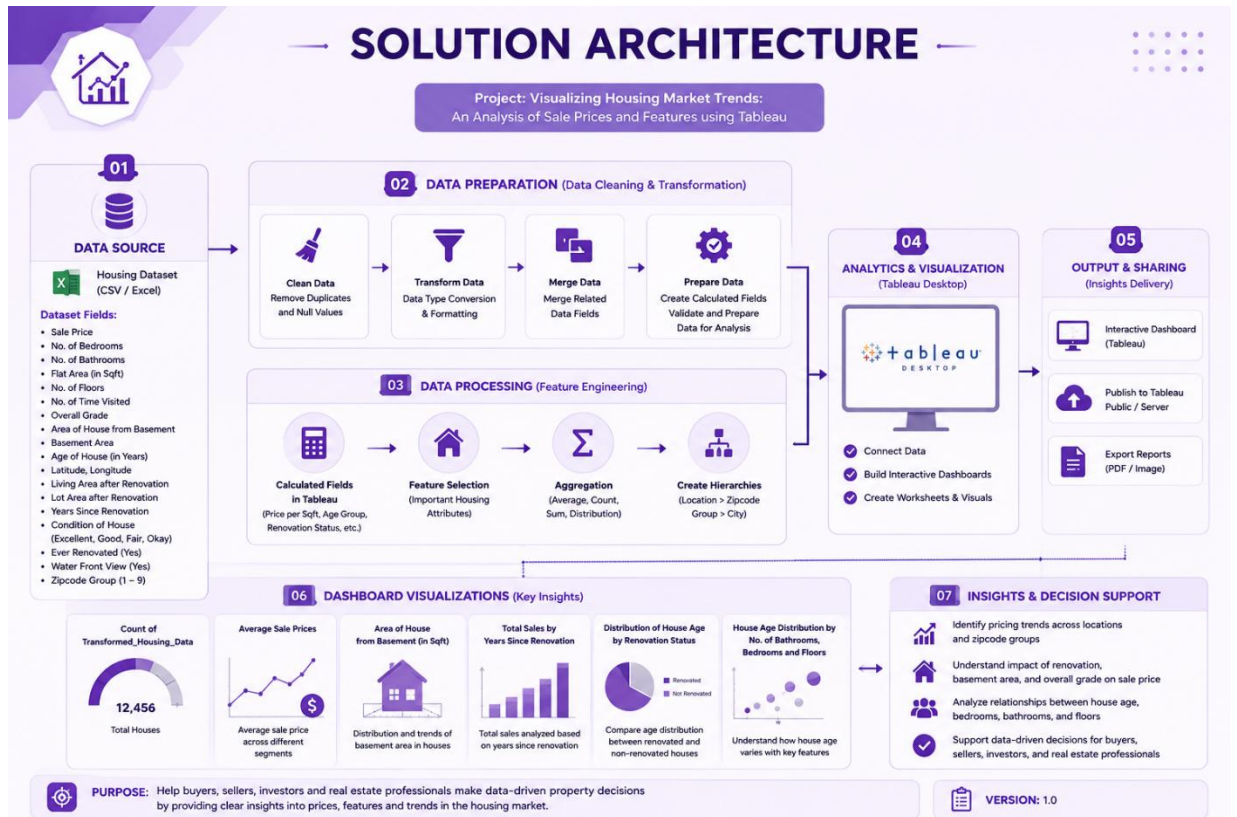
4.3 Solution Architecture Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

Solution Architecture:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behaviour, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.

- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram:



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection & Extraction from Database	USN-1	As a Student, I can dataset was downloaded from the Kaggle website, Extracted and then loaded the dataset into tableau.	2	High	Rahul Kumawat
Sprint-1	Data Preparation	USN-2	As a Student, I can Data Cleaning and transform data Field Renaming and Final Formatting Prepare data for visualization.	3	High	Rahul Kumawat
Sprint-1	Data Visualization	USN-3	As a Student, I can view the Count of Transformed Housing Data and the Average Sale Prices.	3	High	Rahul Kumawat
Sprint-1	Data Visualization	USN-4	As a Student, I can Area of House from Basement (in sqft.) and Total Sales by Years Since Renovation.	3	High	Rahul Kumawat
Sprint-1	Dashboard Visualization and Story	USN-5	As a Student, I can Comprehensive House Price Analysis Dashboard Visualizing Market Trends: An Analysis of Sale Prices and Features using Tableau Story	3	High	Rahul Kumar Gurjer
Sprint-2	Data Performance Testing	USN-6	As a Student, I can Check Number in Visualization and Graphs Utilization of Filters.	5	High	Rahul Kumawat
Sprint-2	Trend Analysis	USN-7	As a Student, I can compare sale prices with house features to identify market trends.	5	Medium	Rahul Kumawat

Sprint-2	Renovation Analysis	USN-8	As a Student, I can analyse total sales by years since renovation.	3	Medium	Rahul Kumawat
Sprint-2	Report Analysis	USN-9	As a Student, I can use dashboard Visualization insights to support housing market decisions.	5	Medium	Rahul Kumar Gurjer
Sprint-2	Publishing	USN-10	Publish Proper Project File in Tableau.	2	Medium	Rahul Kumawat

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	14	3 Days	1 July 2026	3 July 2026	14	3 July 2026
Sprint-2	20	3 Days	4 July 2026	6 July 2026	20	6 July 2026

Velocity:

Total Story Points:

Sprint 1 = 14

Sprint 2 = 20

Velocity= Total Story Points Completed/ Number of Sprints

Total story Points = 14+20=34

No. of Sprints = 2

Velocity = 34/2

17(Story Points per Sprint)

My team's velocity is 17 Story Points per Sprint.

Burndown Chart:

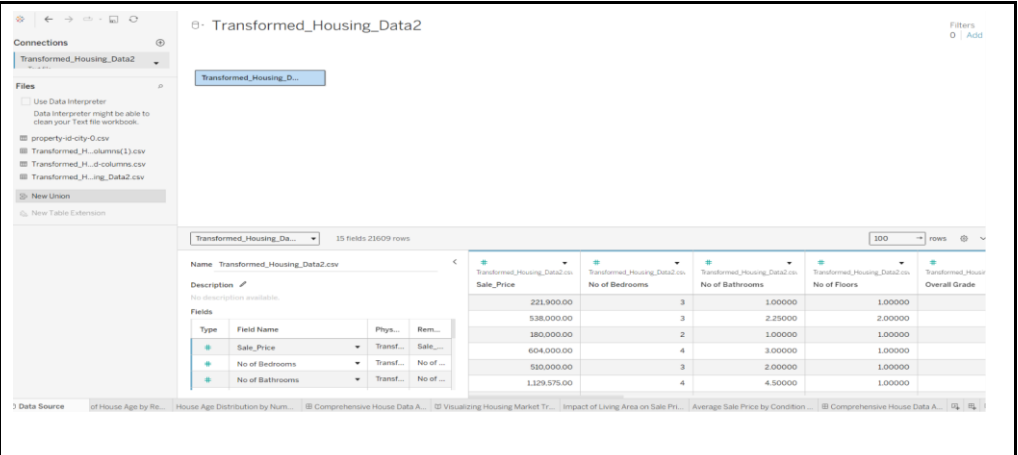
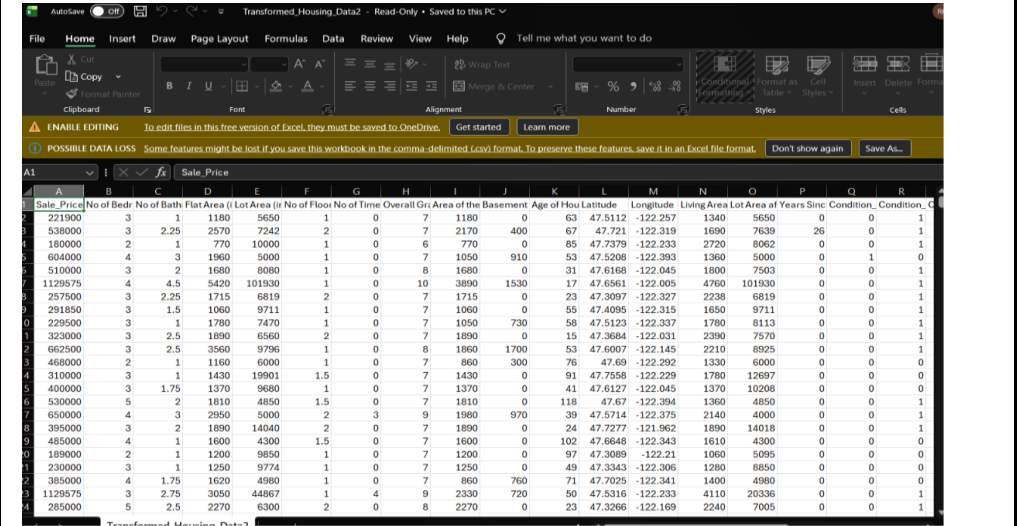
A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Model Performance Testing:

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	The housing dataset was successfully uploaded and rendered in Tableau Public. All dataset fields, including Sale Price, Bedrooms, Bathrooms, Flat Area, Floors, Basement Area, House Age, Renovation Status, Overall Grade, were loaded correctly and made available for further analysis.

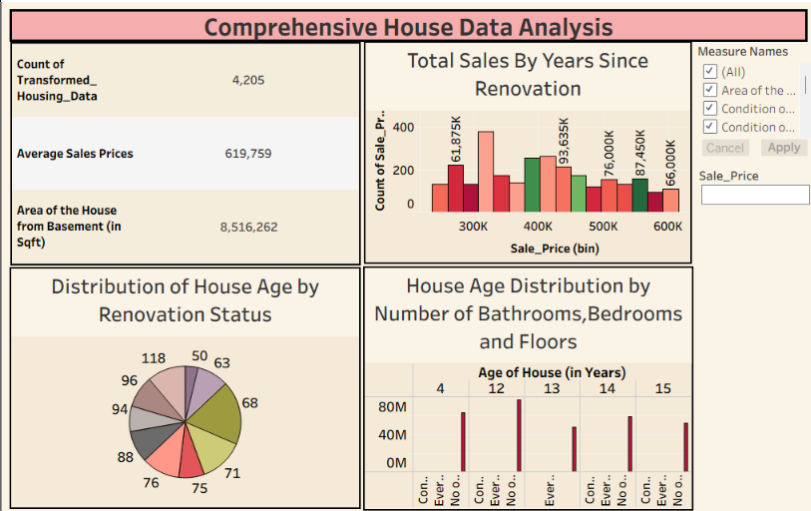
		
2.	Data Preprocessing	Removed null values, verified data types, created calculated fields, grouped Zipcode (1–9), validated renovation and condition fields before visualization. 
3.	Utilization of Filters	Filters used: Sales Price, Years Since Renovation, Number of Bedrooms, Number of Bathrooms, Number of Floors, House Condition, Ever Renovated Yes, Since Renovation Year and Waterfront View. Filters used across different worksheets: Year of Dates (2019 vs 2020), Month of Dates (January to December), Regions (ER, NER, NR, SR, WR), Top N, and Bottom N parameters. Aggregation functions: SUM(Usage) was primarily used for total load mappings, and AVG(Usage) was applied to monitor ongoing baseline shifts. Worksheets utilize coordinated dynamic action filters to allow synchronized dashboard cross-filtering.
4.	Calculation fields Used	IN this Visualization (Total Sales by Years Since Renovation) Use Create Calculation field and other visualization normal calculation fields Use. Logic / Syntax: Mapped via custom integer parameters linked with rank/index functions (INDEX()) to dynamically filter states based on the selection.

Purpose: Solves Tableau's order of operations conflict, allowing the dashboard to dynamically rank the top or bottom states within specific geographical region filters without breaking.

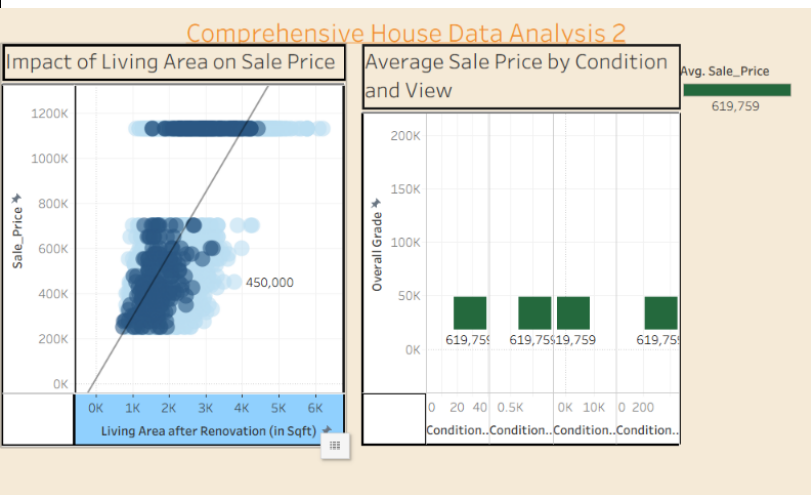
5. Dashboard design

- 1. Count of Transformed Housing Data.
- 2. Average Sale Prices
- 3. Area of House from Basement (Sqft)
- 4. Total Sales by Years Since Renovation
- 5. Distribution of House Age by Renovation Status
- 6. House Age Distribution by Bathrooms, Bedrooms and Floors.
- 7. Impact of Living Area on Sale Price
- 8. Average Sale Price by Condition and View

All Visualization in use Dashboard Design (Comprehensive House Price Analysis)



f.g.->Dashboard 1



		f.g->Dashboard 2
6.	Story Design	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau Story

7. RESULTS

7.1 Output Screenshots

Visualizations:

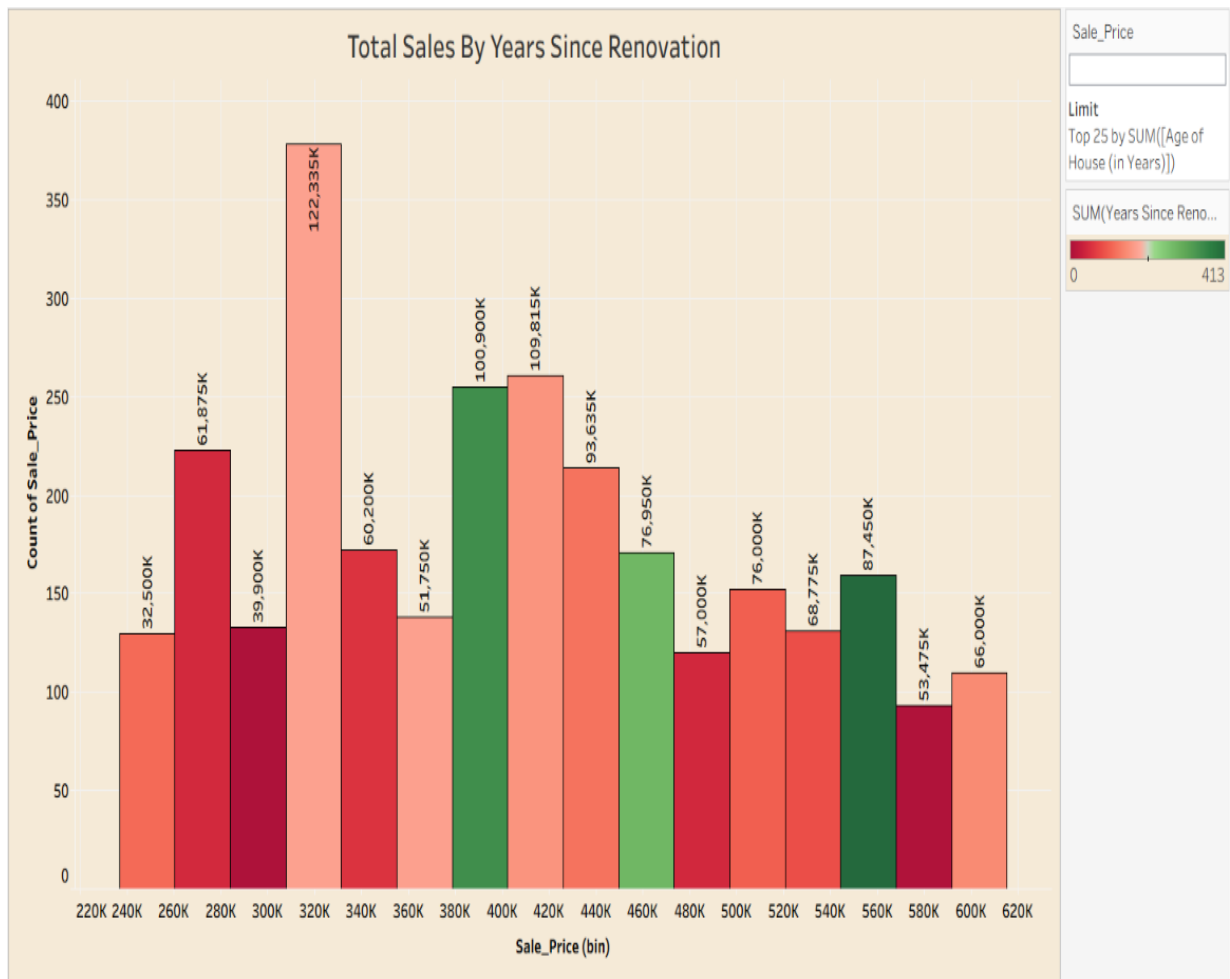
Count of Transformed Housing Data

Average Sale Prices

Area of House from Basement (in Sqft)

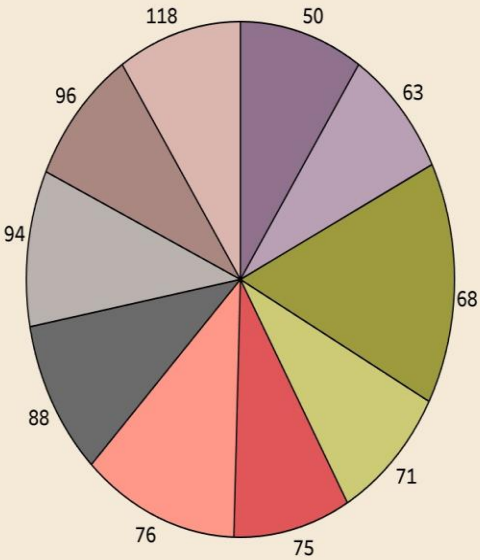
Count of Transformed_Housing_Data	4,205
Average Sales Prices	619,759
Area of the House from Basement (in Sqft)	8,516,262

Total Sales by Years Since Renovation



Distribution of House Age by Renovation Status

Distribution of House Age by Renovation Status



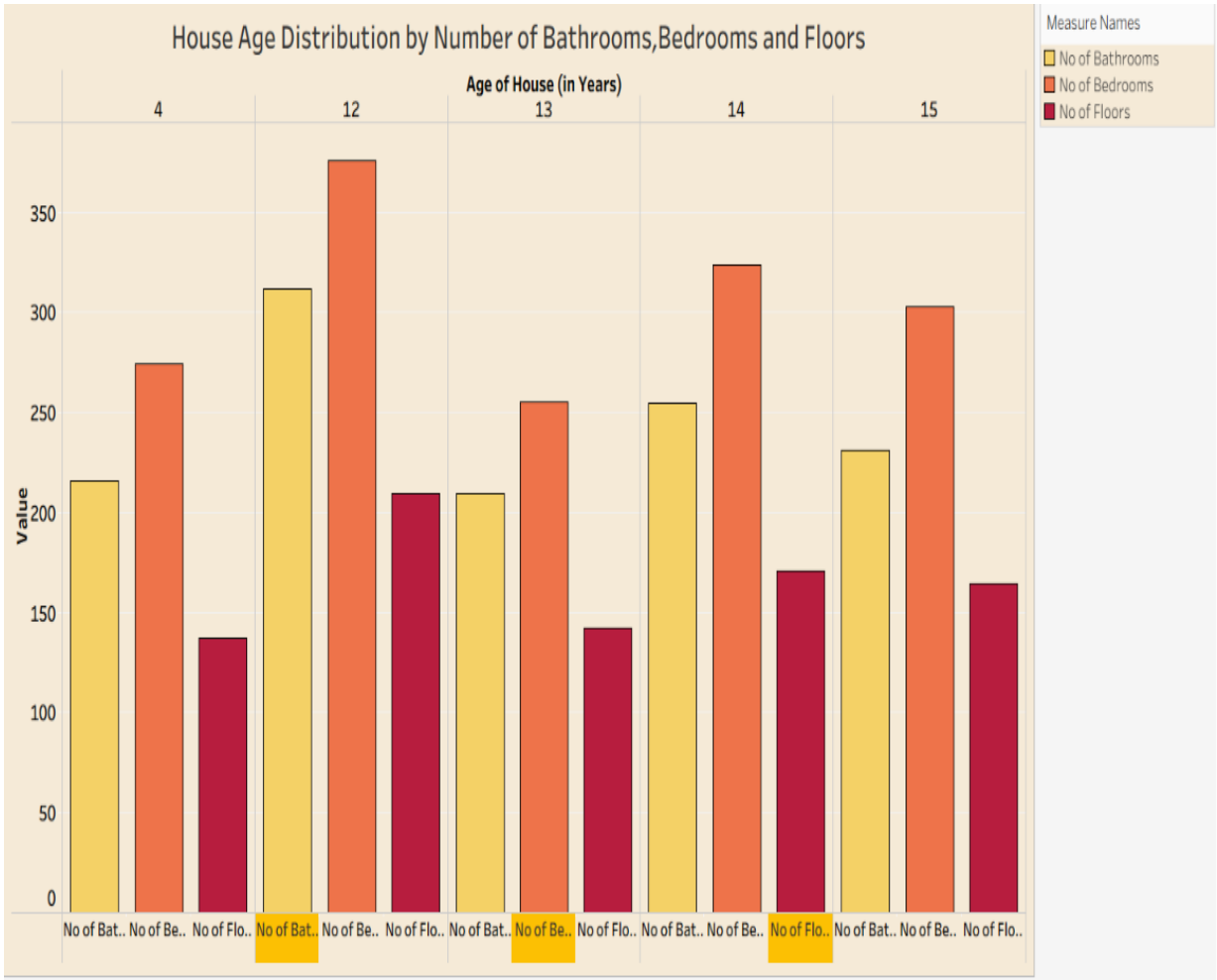
Age of House (in Years)

- 50
- 63
- 68
- 71
- 75
- 76
- 88
- 94
- 96
- 118

SUM(Ever Renovated Y...

204

House Age Distribution by Number of Bathrooms, Bedrooms, and Floors

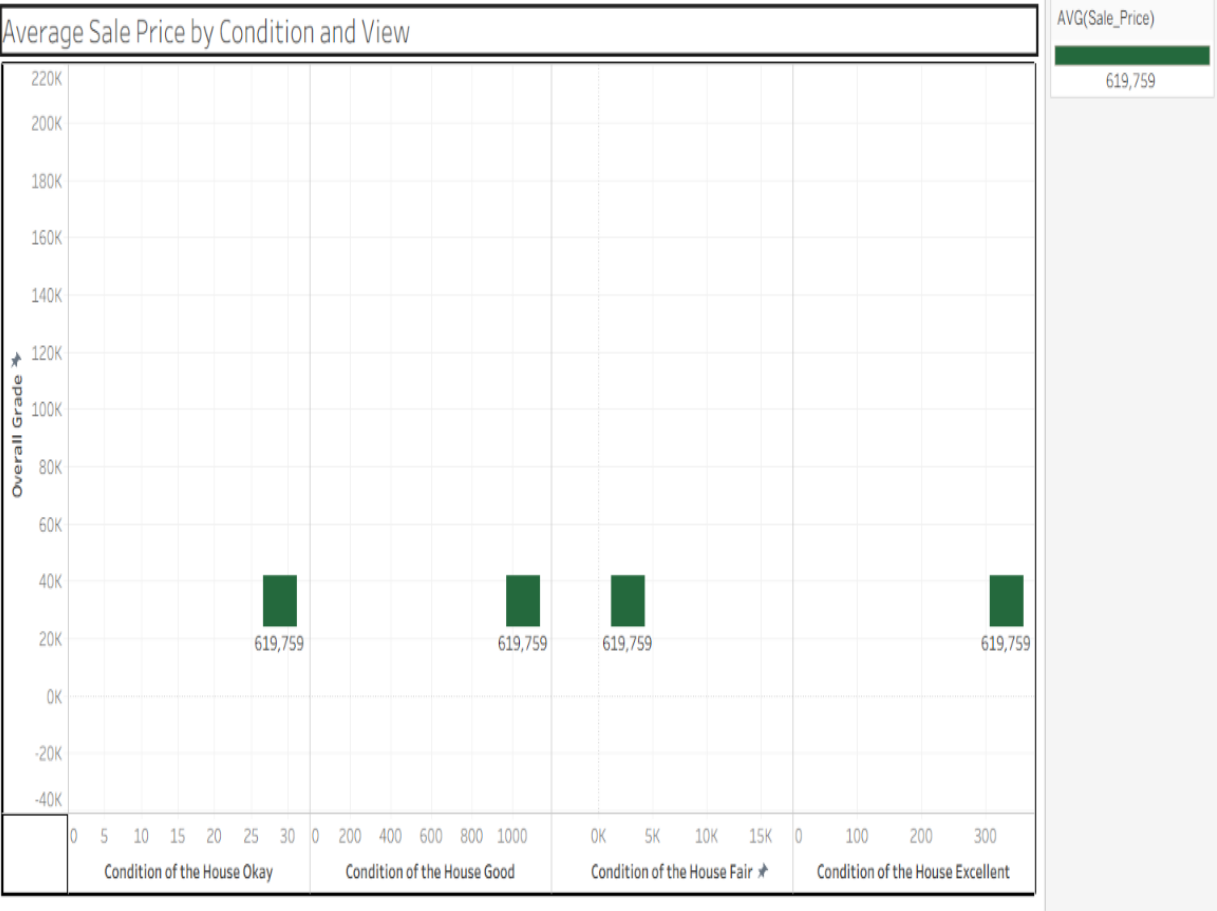


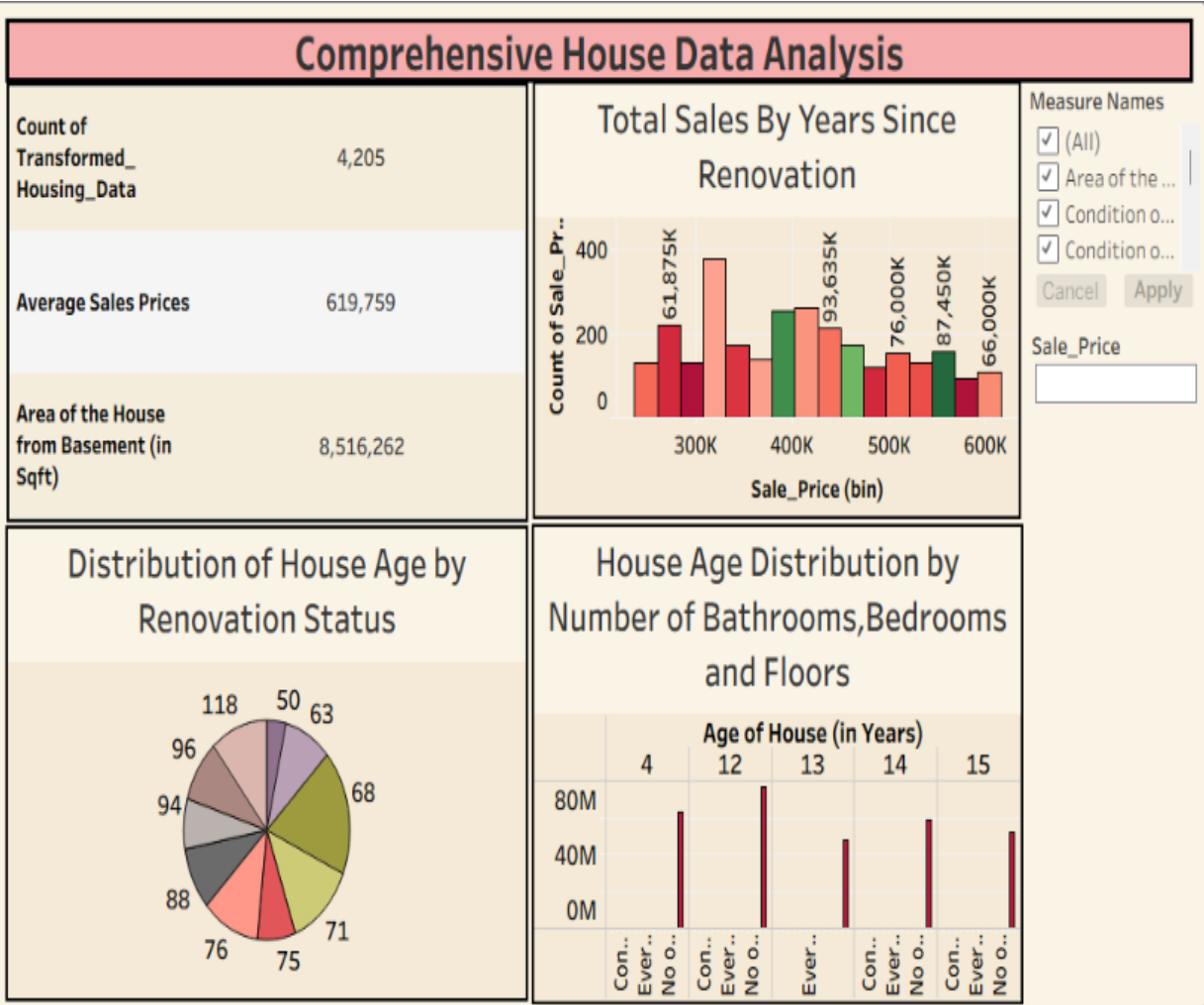
Impact of Living Area on Sale Price

Impact of Living Area on Sale Price



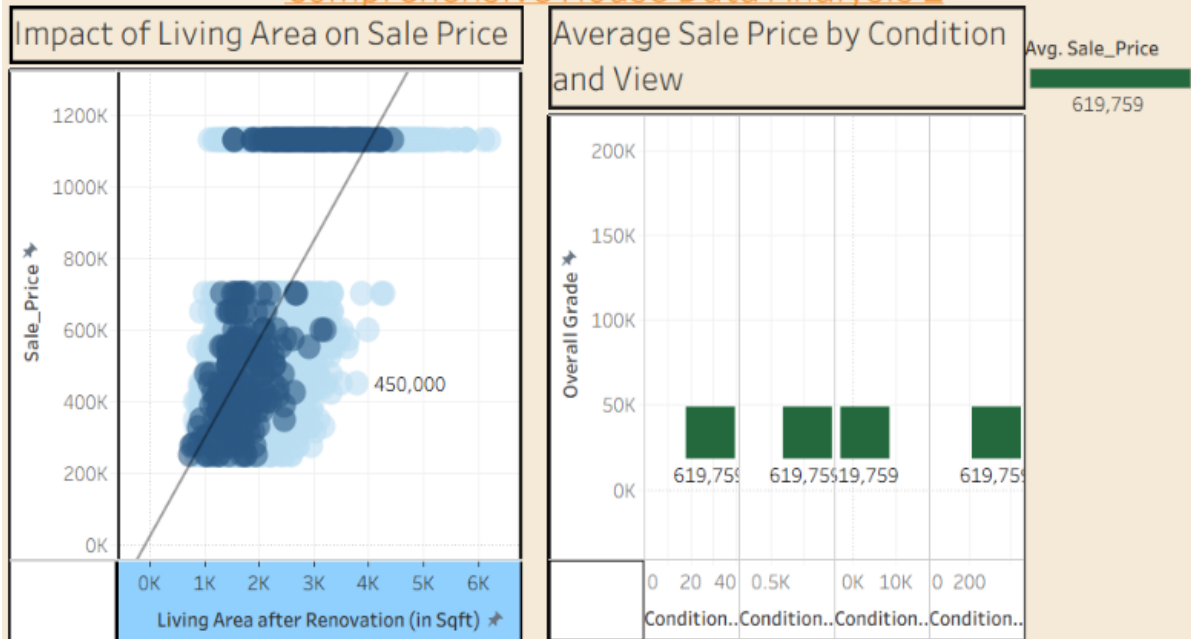
Average Sale Price by Condition and View





Comprehensive House Data Analysis 2

Comprehensive House Data Analysis 2



Story:

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau Story

1. Total Sales By Years Since Renovation

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau Story

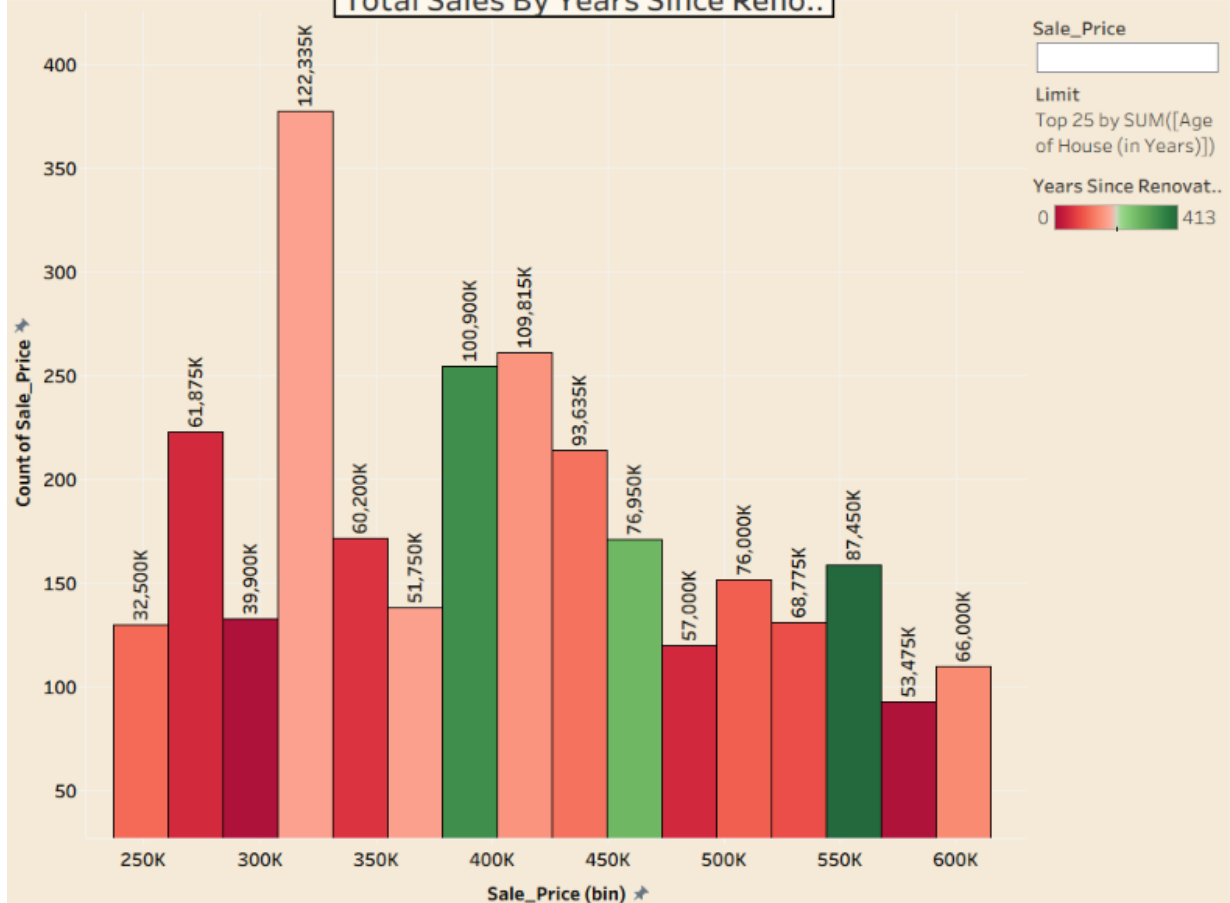
House age distribution is more influenced by the number of

The age of houses varies based on their renovation status, providing

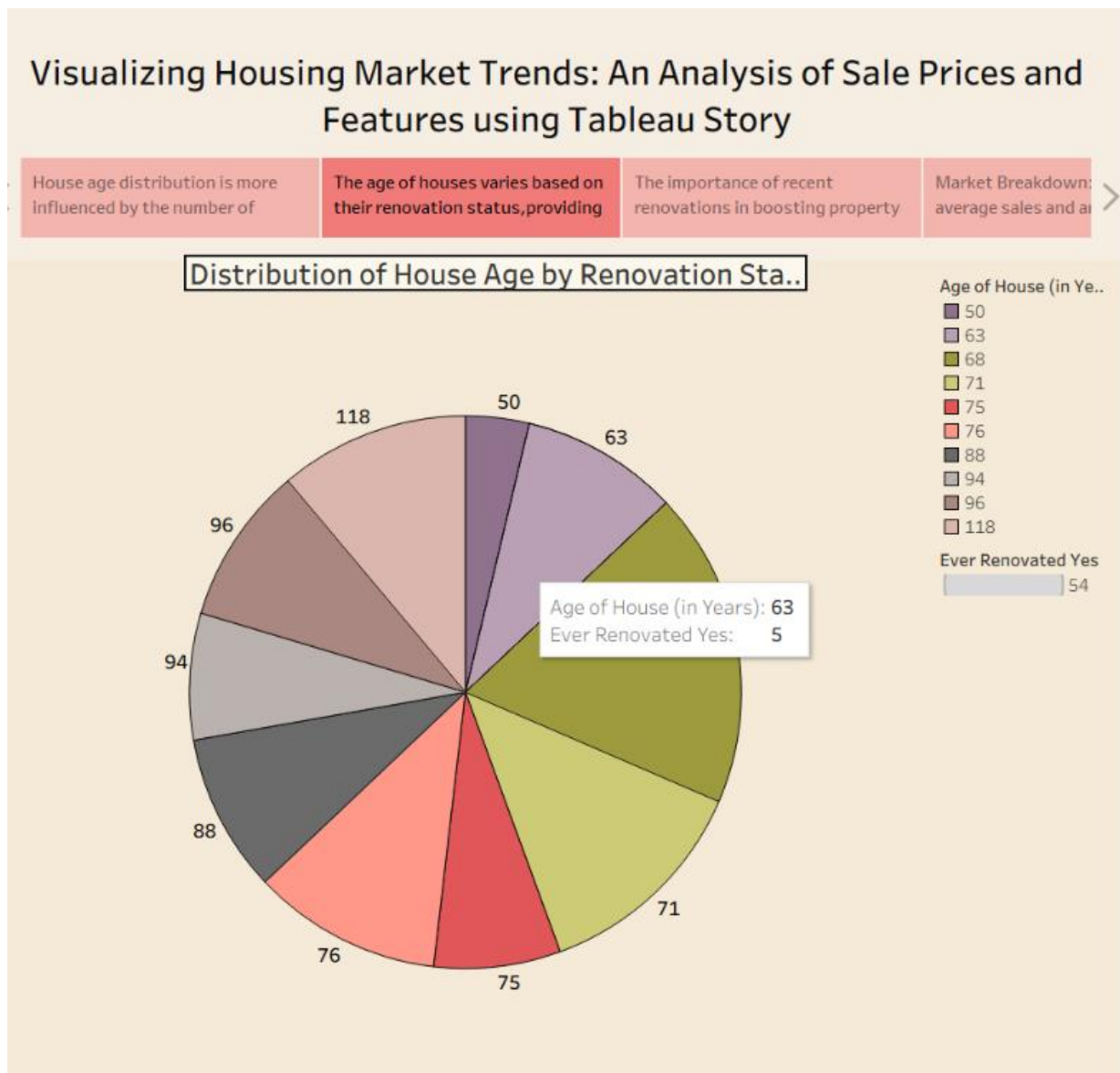
The importance of recent renovations in boosting property

Market Breakdown: average sales and ai

Total Sales By Years Since Reno..



2. Distribution of House Age by Renovation Status



3. House Age Distribution by Number of Bathrooms, Bedrooms and Floors

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau Story

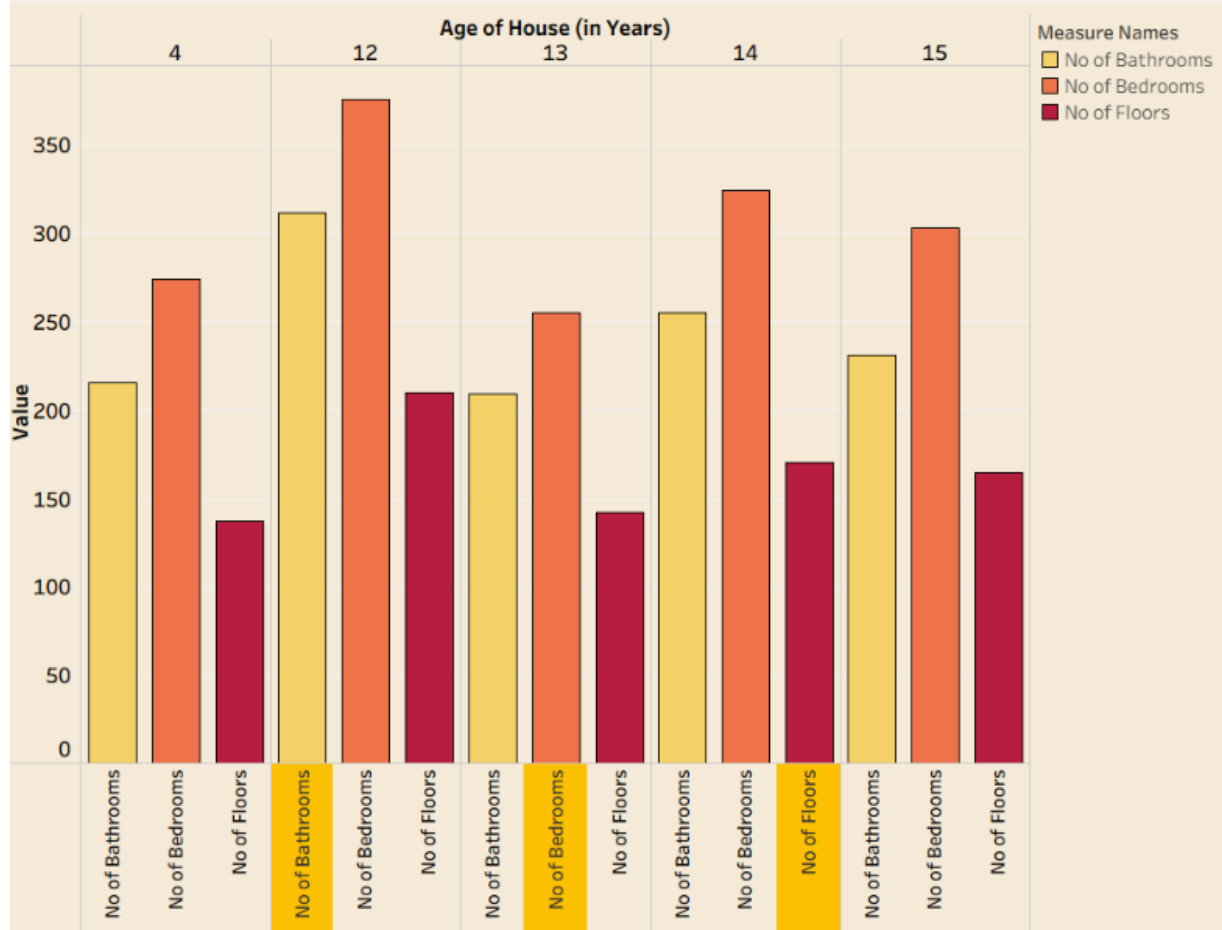
House age distribution is more influenced by the number of

The age of houses varies based on their renovation status, providing

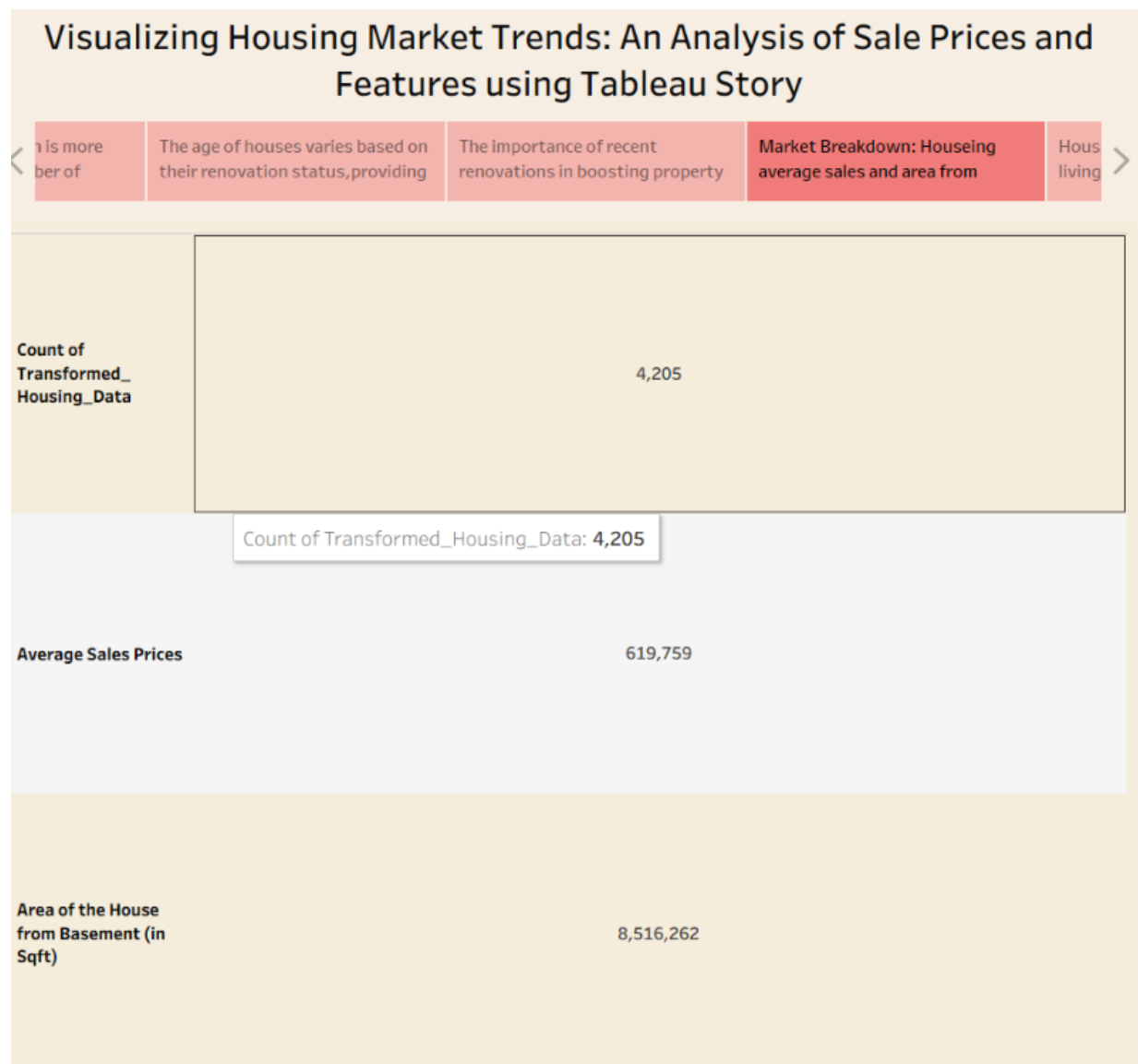
The importance of recent renovations in boosting property

Market Breakdown: average sales and a

House Age Distribution by Number of Bathrooms, Bedrooms a..

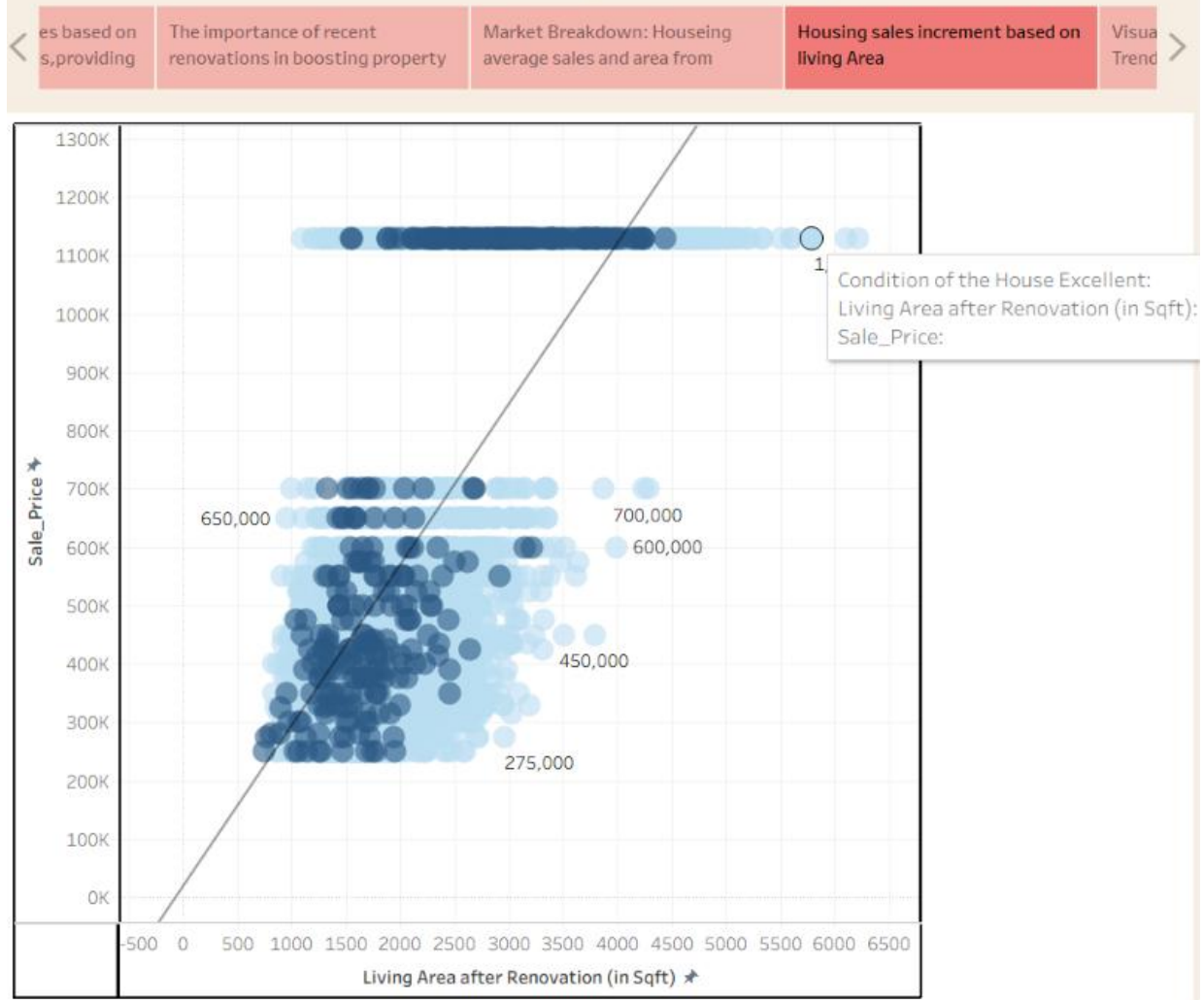


4. Visualization



6. Impact of Living Area on Sale Price

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau Story



6. Average Sale Price by Condition and View

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau Story



Here is Tableau Link →

<https://public.tableau.com/app/profile/rahul.kumawat1942/vizzes>

8. ADVANTAGES & DISADVANTAGES

1. Advantages

- * Helps buyers understand how factors such as location, house size, number of bedrooms, bathrooms, and overall condition influence sale prices.
- * Assists sellers in determining competitive property prices based on market trends and property features.

- * Enables real estate professionals to identify pricing patterns and evaluate housing market performance.
- * Highlights the impact of renovations, house age, and structural features on property value.
- * Supports investors in identifying areas and property types with better investment potential.
- * Presents housing information in an easy-to-understand visual format, enabling quicker and more informed decision-making.
- * Provides comparative insights across different zip code groups and property characteristics, supporting market evaluation.

2.Disadvantages

- * The accuracy of the insights depends on the quality and completeness of the housing dataset.
- * The project does not consider external economic factors such as interest rates, inflation, government housing policies, or market demand.
- * Property prices may change over time, making historical data less representative of current market conditions.
- * The analysis is limited to the variables available in the dataset and may not include all factors affecting house prices.
- * Local market conditions and buyer preferences may influence property values beyond the scope of this project.

9. CONCLUSION

The “Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau” project provides valuable insights into the factors that influence residential property prices. By examining property characteristics such as house area, number of bedrooms and bathrooms, renovation history, house age, location, and overall condition, the project helps stakeholders better understand housing market trends.

The project supports informed decision-making for home buyers, sellers, real estate professionals, and investors by presenting clear information on pricing patterns and property features. It also demonstrates how multiple housing attributes contribute to property valuation and market performance.

Overall, this project serves as a practical decision-support system that improves transparency in the housing market and enables stakeholders to make well-informed real estate decisions based on reliable housing information.

10. FUTURE SCOPE

The project can be further enhanced by incorporating additional data sources and advanced analytical techniques to provide more comprehensive housing market insights. Future enhancements include:

- * Integrating real-time property listings and current market prices for up-to-date analysis.
- * Including economic indicators such as interest rates, inflation, and mortgage trends to better understand their impact on housing prices.
- * Adding neighbourhood facilities such as schools, hospitals, transportation, shopping centres, and parks to evaluate location advantages.
- * Developing predictive models to estimate future house sale prices based on property features and market conditions.
- * Incorporating crime rates, environmental factors, and population growth for more comprehensive property evaluation.
- * Creating personalized property recommendation systems based on user preferences and budget.
- * Expanding the project to compare housing market trends across multiple cities or regions.
- * Developing a mobile-friendly interface to provide easy access to housing market insights for buyers, sellers, and real estate professionals.

These enhancements would make the project a more powerful decision-support solution, helping stakeholders better understand market dynamics, evaluate property values, and make informed investment and purchasing decisions in the real estate sector.

11. APPENDIX

Dataset Link:

<https://www.kaggle.com/datasets/rituparnaghosh18/transformed-housing-data-2>

Tableau Public Link:

https://public.tableau.com/app/profile/rahul.kumawat1942/viz/VisualizingHousingMarketTrends_17830659554540/HouseAgeDistributionbyNumberOfBathroomsBedroomsandFloors

[GitHub & Project Demo Link :](#)

GitHub:

<https://github.com/rkumawat64214-sys/Visualizing-Housing-Market-Trends>

Project Demo Link:

[explanation Video for project -Visualizing Housing Market Trends.mp4](#)